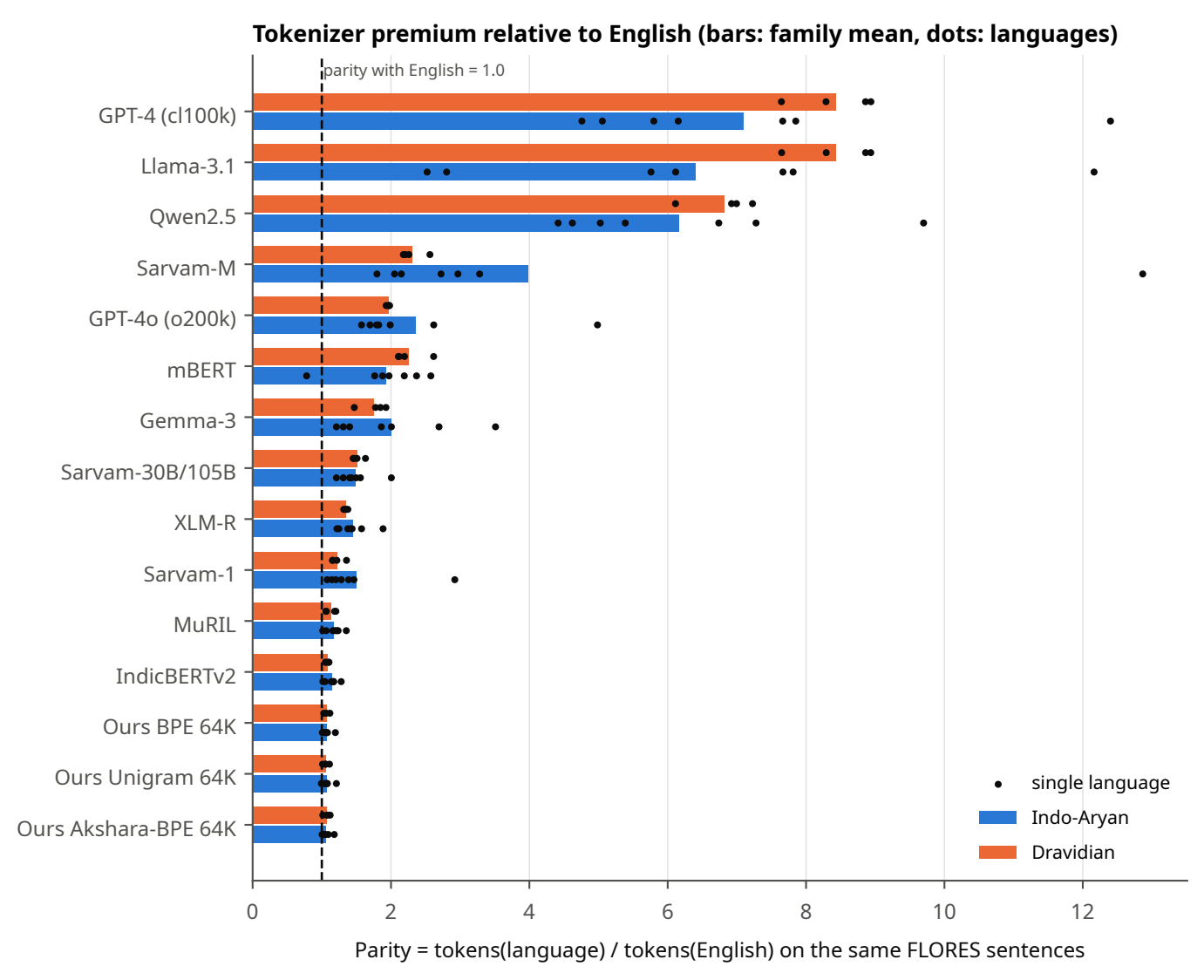
Tokenizer premium relative to English (bars: family mean, dots: languages)
parity with English = 1.0
GPT-4 (cl100k)
Llama-3.1
Qwen2.5
Sarvam-M
GPT-4o (o200k)
mBERT
Gemma-3
Sarvam-30B/105B
XLM-R
Sarvam-1
MuRIL
IndicBERTv2
Ours BPE 64K
Ours Unigram 64K
Ours Akshara-BPE 64K
single language
Indo-Aryan
Dravidian
0
2
4
6
8
10
12
Parity = tokens(language) / tokens(English) on the same FLORES sentences